

Shreshth Singh

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Summary — Skilled developer who turns outdated systems into sleek, modern solutions. Expert in simplifying complex code and applying the latest development practices.

Skills

Technologies React, NEXTJS, ExpressJS, NodeJS, MongoDB, Tailwind CSS, Material-UI

Languages Python, Java, JavaScript, C

Experience

Teaching Assistant — Sardar Patel Institute of Technology (SPIT), Mumbai

Jan 2025– Present

Mumbai, India

- Academic management, streamlining operations and reducing processing time .
- Supervised 15+ undergraduate lab sessions, ensuring adherence to safety protocols and smooth experiment execution.
- Mentored students on practical curriculum components, resolving technical doubts and enhancing conceptual understanding.
- Automated grading workflows for lab reports and assignments, improving evaluation efficiency .
- Facilitated coordination between faculty and students, boosting lab operation efficiency and communication.

Projects

PathfinderAI AI Career Recommendation Platform (Event-Driven Chat System)

[GitHub Link](#)

(Next.js, Inngest, Supabase, Google AI, Clerk, Background Processing)

- Developed AI-powered career guidance platform with Next.js and Google Gemini AI, implementing event-driven workflows using Inngest for background processing of user interactions and AI response generation.
- Built scalable background job system with Inngest for handling career assessments, user onboarding events, and batch processing operations while maintaining real-time chat functionality and session persistence.
- Integrated Supabase PostgreSQL database with secure authentication, created RESTful API endpoints, and implemented modern UI patterns with responsive design for optimal user experience.

RealityNet (Deep Learning-Based Deepfake Detection with Active Learning)

[Demo Link](#)

(PyTorch, Computer Vision, Statistical Analysis, Feature Engineering)

- Architected Meso4 CNN with 94%+ accuracy for deepfake detection using PyTorch, achieving 1s inference time on production deployment
- Built interactive Streamlit web application with real-time image classification
- Implemented active learning pipeline with SQLite integration for continuous model improvement and statistical performance tracking
- Deployed production-ready solution on Streamlit Cloud with automated CI/CD, dependency management, and scalable architecture design

Book Recommendation System

[Demo Link](#)

(Streamlit, Collaborative Filtering, Data Analysis)

- Built a Streamlit-based personalized book recommender using collaborative filtering on 1.1M+ user ratings and 270K+ books.
- Designed an interactive UI with search, rating, reading history, and an analytics dashboard for user insights.
- Optimized performance with model caching, data preprocessing, and external API integration for book cover visuals.

Achievements

- Solved **800+ coding problems** across platforms like LeetCode, Codeforces, and GeeksforGeeks, showcasing strong algorithmic skills.
- Ranked **2989 out of 24,518** participants in LeetCode **Weekly Contest 445**, placing in the top 12% globally.
- Among the **top 15 teams** in the **Nomura KakushIN GenAI Hackathon**, building an innovative AI-driven solution under strict model and time constraints.

Education

Sardar Patel Institute of Technology

Bachelor of Technology in Computer Science with Specialization in Data Science